

19L720 - PROJECT WORK I

DESIGN AND ANALYSIS OF CMOS BASED HYBRID LOW NOISE AMPLIFIER FOR X-BAND RADAR APPLICATION

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Dissertation submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF ENGINEERING

Branch: ELECTRONICS AND COMMUNICATION ENGINEERING

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION
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PSG COLLEGE OF TECHNOLOGY

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SYNOPSIS

This project presents the design and simulation of a hybrid six-stage Low Noise Amplifier (LNA) operating at 9.5 GHz using 45 nm CMOS technology, intended for X-band radar and satellite communication applications. The work aims to achieve an optimal balance between gain, noise figure (NF), stability, and power efficiency, overcoming the limitations traditionally associated with CMOS-based RF amplifiers at microwave frequencies. The proposed architecture integrates four single-ended noise-optimized stages followed by two differential gain stages interconnected through on-chip transformers, thereby achieving both high gain and excellent noise performance in a compact and integrable structure.

The initial single-ended stages employ inductive source degeneration to minimize noise figure while maintaining good input matching with a $50\ \Omega$ source. Subsequent differential stages utilize transformer coupling to provide single-ended-to-differential signal conversion, DC isolation, and impedance transformation. The transformers also contribute to broadband matching and improved stability by reducing the inter-stage loading effect. The circuit was designed and simulated using Cadence Virtuoso and SpectreRF environments with the gpdk045 PDK, incorporating accurate device models for 45 nm CMOS. Analytical design equations were applied to determine the transistor dimensions, bias currents, and passive element values required for resonance at the target frequency.

Simulation results confirm that the hybrid LNA achieves a gain of approximately 32 dB, input return loss (S_{11}) and a noise figure of 4.2 dB for the complete six-stage amplifier. The total power consumption is approximately 12 mW, making it suitable for low-power radar and wireless front-end systems.

The proposed design demonstrates that high-performance X-band LNAs can be realized using deep-submicron CMOS processes, providing a low-cost and fully integrable alternative to traditional GaAs and SiGe amplifiers. Through the hybrid approach and transformer-based coupling, the design effectively addresses the trade-offs among gain, noise, and power consumption. The methodology developed in this project can serve as a foundation for future work involving complete receiver front-ends, adaptive bias control, and scaling toward millimetre-wave applications beyond 60 GHz.

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CHAPTER 1

INTRODUCTION

A Low Noise Amplifier (LNA) is a vital component in the front end of any radio frequency (RF) receiver system. Positioned immediately after the antenna, the LNA amplifies very weak incoming signals while adding as little noise as possible. Its primary purpose is to enhance the signal-to-noise ratio (SNR), ensuring that the useful signal is strong enough for further processing in subsequent stages such as mixers and intermediate-frequency amplifiers.

Since the LNA is the first active stage in the receiver chain, its performance has a direct impact on the overall sensitivity, selectivity, and dynamic range of the system. A poorly designed LNA can significantly degrade the SNR, leading to loss of weak signals that cannot be recovered later. Therefore, achieving high gain with a low noise figure, along with impedance matching and stability, is critical in LNA design.

Modern communication and radar systems rely heavily on LNAs to ensure reliable signal detection under low-power or long-distance conditions, making them one of the most essential building blocks in RF front-end design.

1.1 ROLE OF LNA IN RADAR SYSTEM

In a radar system, the Low Noise Amplifier (LNA) plays a crucial role in the receiver front-end, where it amplifies the weak echo signals reflected from the target while adding as little noise as possible. The primary objective of the LNA is to maximize the signal-to-noise ratio (SNR) of the received signal so that subsequent stages, such as mixers and analog-to-digital converters, can process it effectively.

Since radar systems often operate over long ranges and receive signals that are several orders of magnitude weaker than the transmitted pulse, the LNA's low noise figure and high gain are vital for ensuring detection sensitivity. The amplifier must also maintain linearity to preserve the integrity of the radar return and impedance matching to minimize reflection losses between the antenna and receiver circuitry.

An efficient LNA directly influences the overall receiver sensitivity, range resolution, and accuracy of the radar system. In modern radar applications—such as automotive, defense,

and weather monitoring—LNAs are designed to operate over wide bandwidths and under varying environmental conditions while ensuring stable and reliable performance.

1.1.1 OVERVIEW OF RADAR AND X-BAND OPERATION

The X-band frequency range, extending from 8 GHz to 12 GHz, is widely used in radar, satellite, and point-to-point communication systems. In radar receivers, especially those used for weather monitoring, marine navigation, and military surveillance, the detection range and angular resolution depend heavily on receiver sensitivity, which in turn is dictated by the LNA's noise performance. A modest degradation of 1 dB in noise figure can translate to a 10 % or greater reduction in detection range.

Similarly, in satellite and microwave backhaul links, where signals arrive at the receiver with power levels of only a few microvolts, the LNA must provide high gain with minimal distortion. Therefore, for X-band front-ends the amplifier must combine a low noise figure (typically below 4 dB), high gain (15–20 dB), good input matching to 50 Ω , and unconditional stability across the operating band. Achieving all these simultaneously within the limitations of integrated-circuit technology remains a major challenge.

1.2 OVERVIEW OF CMOS TECHNOLOGIES

The 45 nm CMOS process provides a well-balanced trade-off between high speed, low power, and integration density. The short channel length and thin gate oxide yield a high transconductance (g_m) and low parasitic capacitance, extending the transistor's operating range into the tens of gigahertz. A nominal supply voltage of 1 V enables low static power dissipation, an important factor in portable and automotive radar modules. In addition, advanced metal stacks available in this node allow the realization of high-Q spiral inductors and transformers suitable for tuned circuits at 9–10 GHz.

For this project, the gpdk045 CMOS model is adopted within the Cadence Virtuoso design environment. The process offers reliable transistor models, parameterized passive components, and robust support for electromagnetic extraction through Spectre RF simulations, ensuring accurate prediction of circuit behaviour before fabrication.

1.3 TRADE-OFF BETWEEN GAIN, NF, AND POWER CONSUMPTION

The design of a Low-Noise Amplifier (LNA) inherently involves balancing three interdependent parameters — gain, noise figure (NF), and power consumption. Achieving high gain generally requires a large transconductance and increased bias current, which directly increases the LNA's power consumption. However, increasing

transconductance also helps reduce input-referred noise, thereby improving the noise figure. This relationship creates a fundamental design tension where improving one metric may adversely impact another.

In CMOS LNAs, the noise figure is largely determined by the thermal and flicker noise contributions from the input transistor. To minimize noise figure, the transistor must operate in the saturation region with an optimized bias current. The noise optimum occurs when the source impedance seen by the gate equals the transistor's noise-matching impedance. Unfortunately, this condition rarely coincides with the impedance required for maximum power transfer (system match), thus introducing a noise–power matching conflict.

Similarly, the gain of the LNA is directly proportional to the transconductance and the load impedance at the drain terminal. While higher drain resistance or tuned inductors can enhance gain, they may also increase voltage swings, reducing linearity and potentially causing instability. In addition, higher bias currents that increase transconductance also lead to greater DC power dissipation, which is undesirable for portable or battery-powered systems.

Therefore, practical LNA design demands a careful optimization process rather than pure maximization of any single parameter. Designers typically prioritize achieving an acceptable noise figure (for example, less than or equal to 3 dB) while ensuring sufficient gain (15–20 dB) and maintaining power consumption below target limits (for example, 10–15 mW). Techniques such as inductive source degeneration, multi-stage gain distribution, and transformer coupling are often employed to strike a balance among these parameters.

In this work, a two-stage single-ended front-end is used to minimize noise figure and establish impedance matching, followed by four differential gain stages to achieve the required overall gain without excessive power usage. This hybrid configuration ensures that the design achieves high performance with controlled power efficiency, suitable for integration in modern radar and communication receivers.

1.4 PROBLEM DEFINITION

Conventional single-ended LNAs implemented in deep-submicron CMOS struggle to achieve both low noise and high gain under a 1 V supply. Differential amplifiers, while superior in isolation and stability, require precise symmetry and consume larger area and current. Therefore, a need exists for a topology that merges the low-noise capability of single-ended stages with the gain and stability advantages of differential configurations.

The central problem addressed in this project is to design a hybrid LNA, consisting of single-ended and differential amplifiers, that can deliver a power gain of approximately 20–30 dB with a noise figure below 5 dB at 9.5 GHz, while maintaining unconditional stability and low power dissipation within the 45 nm CMOS process constraints.

1.5 OBJECTIVES OF THE PROJECT

The main objective of this research is to design and implement a CMOS-based hybrid Low Noise Amplifier (LNA) operating at 9.5 GHz (X-band) for radar front-end applications. The design aims to achieve a minimal Noise Figure (NF) while maximizing power gain, improving linearity, isolation, and bandwidth through a hybrid topology that combines single-ended and differential stages. It also seeks to realize the amplifier using standard 45 nm CMOS technology and validate its performance through Cadence Virtuoso simulations with layout-level parasitic extraction.

1.6 SCOPE OF THE WORK

The work encompasses the complete circuit-level design, simulation, and validation of the proposed LNA architecture. It begins with transistor-level modelling, proceeds through small-signal analysis and matching network synthesis, and culminates in layout generation and post-layout verification. The design incorporates on-chip spiral inductors for input and load tuning, transformers for inter-stage coupling, and appropriate bias networks to set the transistor operating points.

Although fabrication and measurement are beyond the current project scope, the design follows realistic foundry design rules and includes all parasitic effects extracted from layout. The proposed methodology can therefore serve as a foundation for future implementation and integration into a full RF front-end module.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE SURVEY

2.1.1 A 120 GHz Hybrid Low Noise Amplifier in 40 nm CMOS – IEEE 2024

Cho and colleagues present a six-stage hybrid LNA optimized for sub-THz radar, combining a two-stage single-ended common-source front end with a four-stage differential common-source back end. The hybrid arrangement exploits the inherently low-noise properties of the single-ended input stages while relying on differential stages to deliver high gain and improved rejection of common-mode disturbances. The design incorporates stagger-tuning and carefully optimized matching networks to broaden bandwidth and minimize insertion loss. Measured/simulated performance includes a noise figure (NF) of 5.5 dB, 27.5 dB power gain at 122.5 GHz, an input P1dB of -29.5 dBm, and a 3-dB bandwidth spanning 118–129 GHz, all implemented in 40 nm CMOS with a compact 0.099 mm² core and low power consumption (27.4 mW at 1 V). The authors note that the topology and component scaling are optimized for D-band operation and would require substantial redesign to translate the approach to X-band frequencies; the paper also contains limited discussion on linearity enhancement techniques applicable to lower frequencies.

2.1.2 Ultralow Noise Figure and Broadband CMOS LNA With Three-Winding Transformer and Large Transistor for 5G NR Applications - IEEE T-MTT, 2022

Li et al. explore a broadband LNA architecture targeting sub-6 GHz 5G NR that uses a three-winding transformer to realize broadband impedance matching, DC biasing, and combined noise/gain optimization in a compact topology. The approach pairs wideband transformer matching with large transistor sizing to reduce thermal and flicker noise while preserving linearity. The result is an exceptionally low NF across the band (0.4–0.9 dB) with flat power gain (≈ 19.2 – 20.7 dB), robust input matching ($S_{11} < -10$ dB), and strong linearity (IIP3 between +7.5 and +9.5 dBm), all at modest power (15.5 mW at 1.1 V) in 40 nm CMOS. The work demonstrates an effective route for concurrent NF and bandwidth optimization for sub-6 GHz front-ends, though the large device sizing increases input capacitance and may hamper extension of the same approach to mmWave or higher-frequency bands without redesign.

2.1.3 A D-Band CMOS Linearity-Enhanced Low-Noise Amplifier Using Multiple-Gated Transistors Technique – IEEE 2025

Zhang and Seo present a D-band LNA in 40 nm CMOS that leverages the multiple-gated transistor (MGTR) technique to improve linearity while attempting to retain reasonable noise and gain performance. The topology comprises three fully differential common-source stages with capacitive neutralization and an MGTR-based final stage to mitigate nonlinear distortion. The design achieves a peak gain of 16.5 dB (normal mode) and 14.2 dB (linear mode) around 133 GHz, with NF values of 8.8 dB (normal) and 9.5 dB (linear) at 140 GHz; MGTR operation yields an IIP3 improvement of roughly +5 dB in linear mode. The amplifier attains a wide 3-dB bandwidth (110–145 GHz) at the cost of relatively high NF compared with some state-of-the-art D-band LNAs. The paper highlights MGTR as a practical linearity-enhancement method for sub-THz CMOS but also underscores tradeoffs between linearity and noise that must be addressed for noise-sensitive receiver front-ends.

2.1.4 A 0.024–5 GHz LNTA in 28-nm CMOS for 5G Exploiting Noise Reduction and Current Reuse – IEEE 2020

Chen and co-authors report a wideband low-noise transconductance amplifier (LNTA) implemented in 28 nm CMOS that achieves very broad coverage (24 MHz–5 GHz) using a resistive shunt-feedback common-gate topology augmented with inductive peaking, transformer-based matching, and a noise-canceling loop. This combination yields a flat gain response (≈ 24 dB), low noise figures across bands (≈ 2.2 dB low-band, 3.1 dB high-band), excellent linearity ($\text{IIP3} \approx +13$ dBm), and good input matching ($S_{11} < -10$ dB), while preserving moderate power consumption (22.5 mW at 1.0 V). The design demonstrates effective techniques—current reuse, noise cancellation, and transformer matching—for broadband low-NF operation in CMOS, but it is targeted at sub-6 GHz applications and would need re-engineering for mmWave/D-band/X-band use where device parasitics and scaling constraints differ substantially.

2.1.5 A 7.2–27.3 GHz CMOS LNA With Multistage Noise-Matching Technique – IEEE 2022

This recent CMOS LNA spans 7.2–27.3 GHz using multistage noise-matching to extend coverage down into X-band while keeping noise figure moderate across a wide range. Implemented in an advanced CMOS node, the work reports measured NF in the 3–4 dB range and peak gain roughly in the mid-teens of dB across the band, demonstrating practical wideband noise-matching methods that include stage-by-stage tradeoffs between NF and

gain. The paper is relevant to X-band design because it shows how multistage noise-matching and wideband techniques can be adapted or narrowed for a high-performance single-band 9.5 GHz target, though a narrowband redesign would be needed to push NF lower for a dedicated X-band LNA.

2.2 BASE PAPER

The base paper titled “A 120 GHz Hybrid Low Noise Amplifier in 40 nm CMOS” by Dong Ouk Cho, In Cheol Yoo, Chul Woo Byeon, Dong-Woo Kang, and Bontaek Koo, published in IEEE Access in 2024, presents a six-stage hybrid low noise amplifier (LNA) designed for D-band radar front-end systems operating between 120 and 125 GHz. The architecture integrates two single-ended common-source stages with four differential common-source stages to combine the benefits of both configurations. The single-ended stages at the input are optimized to achieve low noise performance, enhancing signal sensitivity, while the differential stages in the subsequent stages provide high gain, improved linearity, and better common-mode noise rejection. The design employs stagger-tuning to extend bandwidth and uses optimized matching networks to minimize insertion loss across the 118–129 GHz frequency range. The LNA achieves a noise figure of 5.5 dB, a power gain of 27.5 dB at 122.5 GHz, an input P1dB of –29.5 dBm, and a 3-dB bandwidth of 11 GHz. Implemented in 40 nm CMOS technology, it occupies a compact chip area of 0.099 mm² and consumes only 27.4 mW of power at a 1 V supply, offering a well-balanced trade-off between performance, efficiency, and integration density. The hybrid design approach introduced in the paper demonstrates how the combination of single-ended and differential topologies can enhance both noise and gain characteristics, making it a valuable reference for future LNA designs across different frequency ranges. However, as the design is specifically optimized for D-band operation, scaling it to lower frequencies such as the X-band requires modifications to transistor sizing, matching networks, and biasing strategies.

The proposed project builds upon this base paper by redesigning the hybrid LNA for operation at 9.5 GHz (X-band) using 45 nm CMOS technology, with a focus on weather radar applications. The X-band frequency range is widely used in meteorological radar systems because it provides an ideal balance between detection range and resolution, enabling accurate measurement of rainfall intensity, cloud dynamics, and storm structures. In such radar systems, the low noise amplifier plays a crucial role in the receiver front-end as it determines the overall sensitivity and signal-to-noise ratio of the system. The primary goal of this project is to achieve a low noise figure and high power gain to ensure that weak echo signals from atmospheric particles can be effectively detected and amplified for further processing. By adopting the hybrid topology from the base paper and optimizing it for X-band

operation, this project aims to enhance gain, linearity, and bandwidth while minimizing noise and power consumption. The use of transformer-based and spiral inductor matching networks ensures efficient impedance matching across stages, while optimized biasing helps maintain stability and consistent performance. Implementing the design in 45 nm CMOS not only reduces chip size and power usage but also supports large-scale integration, making it suitable for modern compact weather radar systems. Overall, the importance of this project lies in its contribution to improving weather prediction accuracy through enhanced radar receiver performance. A high-gain, low-noise, and energy-efficient LNA directly enhances the radar's ability to detect weak atmospheric signals with greater precision, thereby supporting more reliable and high-resolution weather forecasting applications.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 OVERVIEW OF EXISTING LNA ARCHITECTURE

Low Noise Amplifiers (LNAs) are crucial components in radar receiver front-ends, responsible for amplifying weak input signals while introducing minimal noise. Over the years, CMOS-based LNAs have evolved from single-stage to complex multi-stage architectures to achieve better gain, bandwidth, and linearity. Single-ended designs are known for their simplicity and low power consumption, but they often suffer from limited gain and poor isolation. On the other hand, differential LNAs provide higher gain, improved linearity, and strong common-mode noise rejection, making them suitable for high-performance systems. However, these designs require complex matching networks and occupy larger chip areas.

Advances in multi-stage LNAs introduced cascaded amplifier stages to distribute the total gain and improve bandwidth. Researchers between 2010 and 2020 demonstrated several X-band differential LNAs using 90 nm to 28 nm CMOS technologies that achieved up to 25 dB gain with noise figures below 3 dB. The limitation of such designs lies in their increased area and power consumption due to complex inter-stage matching. To overcome these challenges, transformer-coupled and hybrid architectures were introduced, providing compact impedance transformation and single-ended-to-differential conversion, making them ideal for modern radar systems.

3.2 COMPARISON WITH EXISTING WORK

Recent literature shows a consistent trend of performance improvement in CMOS LNAs through hybrid and transformer-coupled architectures. Early 0.18 μm CMOS LNAs achieved around 10 dB gain with noise figures near 5 dB. In contrast, modern 45 nm designs have reported over 20 dB gain and noise figures below 3 dB. Despite these advancements, challenges remain in maintaining low power consumption, compact layout, and broadband matching at X-band frequencies.

The base paper, “A 120 GHz Hybrid Low Noise Amplifier in 40 nm CMOS”, demonstrated a six-stage hybrid architecture combining single-ended and differential stages through on-chip transformers, achieving 27.5 dB gain and 5.5 dB NF at D-band. However, since the design targets 120 GHz, adapting it to 9.5 GHz (X-band) requires re-optimization of transistor

dimensions, bias points, and passive components due to variations in parasitic behavior at lower frequencies. Our proposed design leverages the same hybrid principle but at X-band, enabling higher inductor quality factor (Q), better impedance matching, and improved gain-to-noise performance at a reduced power budget.

3.3 SURVEY ON HYBRID SINGLE-ENDED AND DIFFERENTIAL DESIGNS

Hybrid architectures merge the low-noise performance of single-ended LNAs with the high gain and linearity of differential stages. Several studies between 2015 and 2024 introduced hybrid topologies where the input section consists of low-noise single-ended amplifiers, followed by differential gain stages coupled through on-chip transformers. The transformer performs dual functions: converting the signal from single-ended to differential mode and acting as a tuned inter-stage matching element. This approach enhances power transfer efficiency and suppresses substrate coupling.

While differential-only LNAs require extensive area for matching networks, hybrid architectures achieve similar or better performance using fewer passive components. Research has shown that these designs can maintain sub-3 dB noise figures with gains exceeding 20 dB. The proposed work adopts this hybrid structure to balance noise performance, linearity, and power efficiency, making it ideal for weather radar front-end design.

3.4 PROPOSED HYBRID LNA DESIGN

The proposed hybrid LNA operates at 9.5 GHz using 45 nm CMOS technology and adopts a six-stage topology: two single-ended input stages followed by four differential gain stages. The first two stages are designed to minimize the noise figure by optimizing transistor dimensions, biasing, and input matching. These stages ensure the amplifier maintains high sensitivity for weak radar echo signals. The subsequent four differential stages amplify the signal further, providing high gain and improving linearity and isolation.

On-chip transformers are used between stages for impedance transformation and single-ended-to-differential conversion. These transformers also serve as tuned loads, improving voltage gain and bandwidth while minimizing insertion loss. The biasing network is carefully designed to ensure stability and temperature independence. The complete design process involves schematic-level optimization, followed by electromagnetic (EM) co-simulation and post-layout verification in Cadence Virtuoso and Spectre RF environments.

The design targets the following performance parameters: a noise figure below 3 dB, a power gain of over 20 dB, an input reflection coefficient (S_{11}) below -10 dB, and power consumption under 30 mW. Such specifications ensure high receiver sensitivity for accurate weather data acquisition in radar systems.

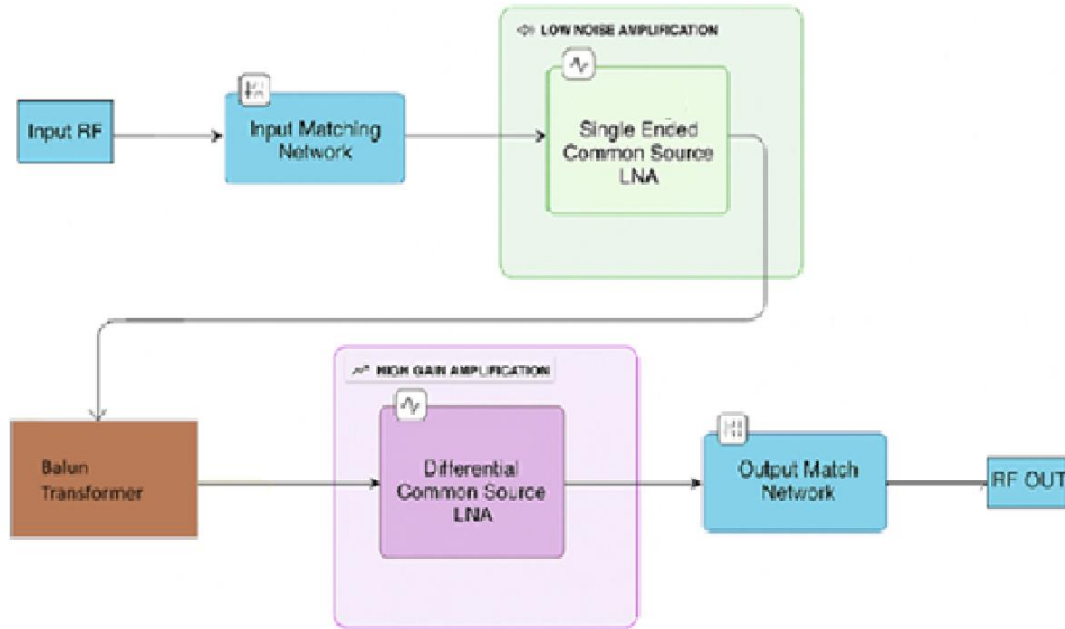


Fig 3.1 block diagram of proposed hybrid LNA

The functional block diagram of the proposed hybrid LNA begins with the RF input from the radar antenna. The first block is the Input Matching Network, composed of spiral inductors and capacitors tuned for $50\ \Omega$ impedance matching, ensuring minimal reflection and maximum signal transfer into the amplifier. This stage also reduces noise figure by providing optimal impedance to the first transistor.

Next, the signal enters the Single-Ended Common-Source LNA Stages, which serve as the low-noise amplification section. These stages provide the initial voltage gain while maintaining low noise contribution. Following this, the amplified single-ended signal is passed to a Balun Transformer, which converts the signal into differential form. The transformer also provides inter-stage impedance matching and acts as a band-selective load, improving both gain and bandwidth.

The differential signal is then fed into Differential Common-Source LNA Stages, where further amplification occurs. These stages enhance the overall power gain, improve linearity, and suppress common-mode noise. The final stage includes an Output Matching Network, which matches the amplifier output to the radar receiver's next block, typically a mixer or filter, ensuring efficient power transfer and flat gain response across the X-band.

This block-level approach ensures that the amplifier delivers high gain, low noise, and robust operation under varying conditions. The hybrid configuration makes the design compact, efficient, and ideally suited for weather radar systems that require high sensitivity and stable signal amplification for accurate weather prediction.

CHAPTER 4

DESIGN OF PROPOSED SYSTEM

This chapter describes the proposed methodology used to design the hybrid Low-Noise Amplifier (LNA) operating at 9.5 GHz using 45 nm CMOS technology. The approach integrates theoretical analysis, small-signal modelling, transistor sizing, bias selection, and the synthesis of matching and coupling networks. The objective is to achieve high power gain, low noise figure, and unconditional stability while maintaining minimal power consumption.

The methodology combines two single-ended amplifier stages optimized for low noise with four differential transformer-coupled stages optimized for gain and stability. The hybrid structure exploits the advantages of both configurations, with transformers performing impedance transformation and single-ended-to-differential signal conversion. The design flow follows a systematic sequence: transistor characterization → bias selection → input matching → load tuning → inter-stage coupling → noise optimization → layout preparation.

4.1 THEORETICAL BASIS OF LNA DESIGN

The design of an LNA is based on the small-signal model of a MOSFET, which captures the transconductance (g_m), gate-source capacitance C_{gs} , gate-drain capacitance C_{gd} , and output resistance (r_o). At high frequencies, parasitic capacitances and inductances dominate impedance behaviour, and matching becomes essential for maximum power transfer.

The voltage gain (A_v) of a single transistor stage is approximated as:

$$A_v = g_m \times R_{load}$$

where G_m is the transconductance and R_{load} is the effective impedance seen at the drain at the operating frequency. For a tuned circuit, R_{Load} is the equivalent parallel resistance of the resonant LC network.

The transducer power gain (G_T), which accounts for source and load matching, is given by:

$$G_T = \frac{|S_{21}|^2(1 - |\Gamma_S|^2)(1 - |\Gamma_L|^2)}{|(1 - S_{11}\Gamma_S)(1 - S_{22}\Gamma_L) - S_{12}S_{21}\Gamma_S\Gamma_L|^2}$$

Here, Γ_S and Γ_L are the reflection coefficients of source and load, respectively. For matched conditions ($\Gamma_S = \Gamma_{in}^*$ and $\Gamma_L = \Gamma_{out}^*$), maximum power transfer occurs.

4.1.1 NOISE FIGURE ANALYSIS

The Noise Figure (NF) quantifies the degradation of signal-to-noise ratio introduced by the amplifier. For a MOSFET in common-source configuration, the noise factor (F) can be approximated as:

$$F = 1 + \frac{\gamma}{\alpha} \left(\frac{g_m}{4R_S} \right) \left| \frac{\Gamma_S - \Gamma_{opt}}{1 + \Gamma_{opt}\Gamma_S} \right|^2 + \frac{R_g + R_s}{R_S}$$

where:

- γ is the channel noise coefficient (typically 2/3 for long-channel, up to 1–2 for deep-submicron MOSFETs),
- α accounts for correlation between drain current and induced gate noise,
- R_S is the source resistance (50 Ω),
- Γ_{opt} is the optimum source reflection coefficient minimizing NF,
- R_g and R_s represent gate and source resistances.

Minimizing $|\Gamma_S - \Gamma_{opt}|$ ensures the lowest noise figure. Since Γ_{opt} depends on device geometry and biasing, transistor sizing and L_S selection (source degeneration) are critical.

The minimum noise figure (NF_{min}) for an LNA is expressed as:

$$NF_{min} = 10 \log_{10} \left(1 + \frac{\gamma}{\alpha} \frac{g_m}{4\pi f C_{gs} R_S} \right)$$

At higher frequencies, G_{gs} and R_S limit noise optimization. Therefore, inductive source degeneration is used to tune the input impedance for simultaneous noise and power matching.

4.1.2 INPUT IMPEDANCE AND MATCHING NETWORK

For a common-source amplifier with inductive source degeneration L_s and gate inductance L_g the input impedance is approximated as:

$$Z_{in} = j\omega(L_g + L_s) + \frac{1}{j\omega C_{gs}} + g_m L_s$$

To achieve simultaneous noise and input matching, L_s and L_g are chosen so that the real part of Z_{in} equals 50 Ω and the imaginary part is cancelled by gate capacitance resonance.

The real part condition:

$$Re(Z_{in}) \approx \omega_T L_s g_m = 50\Omega$$

and the imaginary part condition:

$$\text{Im}(Z_{in}) = \omega(L_g + L_s) - \frac{1}{\omega C_{gs}} = 0$$

From these relations, L_s and L_g can be computed for the given G_m and C_{gs} of the selected transistor.

4.1.3 BIAS POINT AND TRANSISTOR SIZING

Transistor sizing determines both transconductance and parasitic capacitances. The transconductance per unit drain current (g_m/I_D) method is employed to select an operating region that balances gain and linearity.

For the 45 nm CMOS process, $G_m/I_d \approx 15\text{--}20$ S/A corresponds to moderate inversion, yielding optimum noise performance and reasonable headroom. The drain current for each transistor is selected based on desired G_m using:

$$I_D = \frac{g_m}{(g_m/I_D)}$$

The effective width (W) is obtained from the process model relation:

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})$$

where μ_n is carrier mobility, C_{ox} the oxide capacitance per unit area, and V_{th} the threshold voltage. Proper scaling ensures that total C_{gs} remains within limits for the 9.5 GHz target.

4.1.4 DRAIN TUNED LOAD NETWORK

Each amplifier stage employs a tuned LC load resonating at the operating frequency to maximize voltage gain. The drain load inductance L_d resonates with the total drain capacitance $C_{\text{drain_total}}$ which includes the transistor output capacitance, interconnect parasitic, and the transformer's primary capacitance.

The resonance condition is:

$$\omega_0 = \frac{1}{\sqrt{L_d C_{\text{drain_tot}}}}$$

Thus,

$$L_d = \frac{1}{(2\pi f_0)^2 C_{\text{drain_tot}}}$$

For $f_0 = 9.5$ GHz and $C_{\text{drain_total}} \approx 60$ fF, $L_d \approx 4.7$ nH. However, since large inductors are impractical on-chip, smaller values (1–1.5 nH) combined with inter-stage matching capacitors are used to maintain resonance with higher Q and reduced loss.

4.1.5 TRANSFORMER-COUPLED DIFFERENTIAL STAGES

The differential amplifier stages are connected using **on-chip transformers** that serve dual purposes: impedance matching and single-ended-to-differential conversion. The equivalent model of a transformer includes self-inductances L_P , L_S , mutual inductance $M = \sqrt{L_P L_S}$, and series resistances R_P and R_S representing copper losses.

The voltage gain of an ideal transformer-coupled differential stage is given by:

$$A_v = n \times g_m R_{\text{load}}$$

where $n = \sqrt{L_P L_S}$ is the turns ratio. The transformer's coupling coefficient (k) (typically 0.6–0.8 for CMOS spirals) affects bandwidth and insertion loss. The resonance of the secondary circuit is tuned at 9.5 GHz using the secondary inductance and its parallel capacitance.

The biasing method exploits the transformer: VDD is supplied through the primary winding, providing DC isolation while allowing RF coupling. The secondary winding connects to the transistor gates, which are biased via high-value resistors or current mirrors to maintain proper V_{gs} .

CHAPTER 5

IMPLEMENTATION AND VALIDATION

This chapter presents the complete implementation and validation process of the proposed hybrid Low-Noise Amplifier (LNA) in Cadence Virtuoso using the gpdk045 (45 nm CMOS) process.

The design methodology discussed earlier is now translated into practical circuit implementation. Each amplifier stage, passive element, and bias network is carefully modelled, simulated, and tuned to achieve the target operating frequency of 9.5 GHz.

The hybrid LNA consists of two single-ended stages followed by four differential amplifier stages, interconnected through transformer-based coupling networks. Each section was implemented with due consideration to noise performance, gain flatness, impedance matching, and DC biasing integrity. The complete design flow involves schematic design, simulation verification, optimization, and layout with parasitic extraction.

5.1 DESIGN ENVIRONMENT SETUP

The circuit design was carried out in Cadence Virtuoso Analog Design Environment (ADE). The gpdk045 Process Design Kit (PDK) was loaded, providing accurate transistor and passive component models. All simulations were executed using SpectreRF, which supports periodic steady-state (PSS), small-signal (AC/S-parameter), and noise analyses.

The supply voltage (VDD) was fixed at 1.0 V, with bias currents set between 1–3 mA per transistor depending on stage requirements. Simulation frequency sweeps were performed across 8 GHz to 12 GHz to cover the entire X-band. The circuit environment included a 50 Ω source and 50 Ω load for standard matching verification.

5.2 IMPLEMENTATION OF THE SINGLE-ENDED STAGES

The first two stages and last two stages are single-ended inductively degenerated amplifiers, optimized primarily for low noise figure and good input return loss (S_{11}).

5.2.1 INPUT MATCHING NETWORK

The input matching network employs a gate inductance L_g , a source degeneration inductance L_s , and the transistor gate capacitance C_{gs} to achieve simultaneous noise and impedance matching.

$$Z_{in} = j\omega(L_g + L_s) + \frac{1}{j\omega C_{gs}} + g_m L_s$$

The input impedance equation:

$$\text{Re}(Z_{in}) = 50 \Omega, \quad \text{Im}(Z_{in}) = 0$$

was tuned such that,

For 9.5 GHz operation, the typical design values used were:

- $L_s = 0.22 \text{ nH}$
- $L_g = 0.35 \text{ nH}$
- $C_{gs} \approx 60 \text{ fF}$

These values ensure input resonance near 9.5 GHz and optimal noise matching. A series resistor ($R = 50 \Omega$) was added between the source and signal input to stabilize impedance without degrading NF significantly.

5.2.2 DRAIN LOAD AND BIASING

Each single-ended stage used a drain-tuned load, consisting of $L_d \approx 1.0 \text{ nH}$ and a drain parasitic capacitance $C_d \approx 70 \text{ fF}$, resonating at 9.5 GHz:

$$L_d = \frac{1}{(2\pi f_0)^2 C_d}$$

The biasing was provided by a current mirror circuit generating $V_{GS} \approx 0.5 \text{ V}$ and $I_D \approx 2 \text{ mA}$. DC isolation capacitors (C_c) of 200 fF were used between stages. These stages collectively achieved a gain of $\approx 32 \text{ dB}$ and noise figure $\approx 4.2 \text{ dB}$.

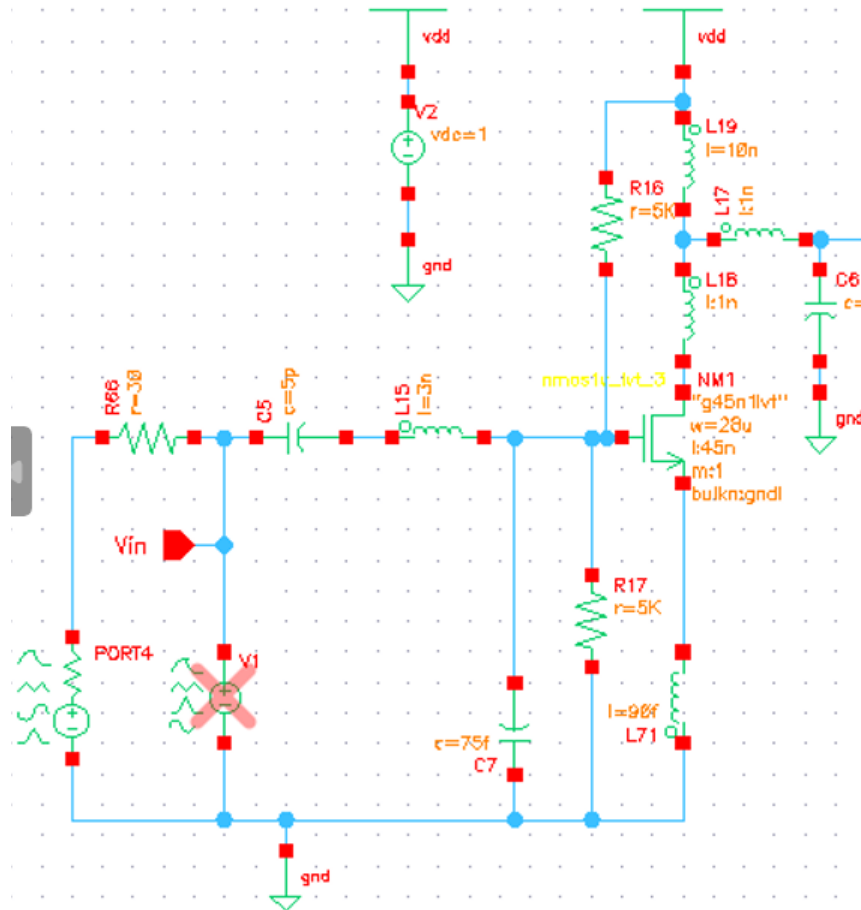


Fig 5.1 Schematic of a Single Stage single ended Amplifier

5.3 IMPLEMENTATION OF DIFFERENTIAL AMPLIFIER STAGES

The next two stages are differential amplifiers coupled through on-chip transformers. Each stage is tuned for maximum midband gain and stable operation.

5.3.1 DIFFERENTIAL PAIR TOPOLOGY

Each differential stage consists of two transistors forming a balanced pair. The transistors are biased symmetrically, with their sources connected to a common tail current source. The load on each drain is provided through transformer windings that resonate at the target frequency.

$$A_v = g_m \times R_{load}$$

where R_{load} is the effective impedance of the tuned transformer secondary at resonance.

5.3.2 TRANSFORMER DESIGN AND MODELLING

The transformer was modelled using coupled inductors L_p and L_s , with mutual inductance $M = k\sqrt{L_p L_s}$, and a coupling coefficient $k = 0.7$.

$$f_0 = \frac{1}{2\pi\sqrt{L_p C_p}} = \frac{1}{2\pi\sqrt{L_s C_s}}$$

The resonance condition for each winding is:

Typical design values:

- $L_p = 0.7 \text{ nH}$, $L_s = 0.6 \text{ nH}$
- $C_p \approx 90 \text{ fF}$, $C_s \approx 100 \text{ fF}$

The primary winding connects to the drain of the preceding stage and is fed through VDD via an RF choke. The secondary winding connects to the next differential pair's gate, providing both AC coupling and DC bias isolation.

5.3 3 BIASING THROUGH TRANSFORMER COUPLING

Each transformer also assists in distributing the bias voltages. The VDD is applied through the primary winding, while the secondary winding carries the gate bias (VGG) of the next stage.

This arrangement provides:

- DC isolation between stages,
- Controlled bias distribution,
- Reduced parasitic coupling, and
- Simpler routing for bias networks.

5.4 BIASING NETWORKS

The circuit uses two separate bias networks:

1. **Gate Bias (VGG) network** – provides a steady bias voltage through large-value resistors ($\geq 100 \text{ k}\Omega$) ensuring negligible RF loading.
2. **Drain Bias (VDD) network** – supplies DC current through RF chokes ($\approx 5 \text{ nH}$) that block RF signals.

Each stage is biased in moderate inversion for linearity and optimum (g_m/I_D) performance. Bypass capacitors of 10 pF are added between VDD and ground to suppress supply ripple.

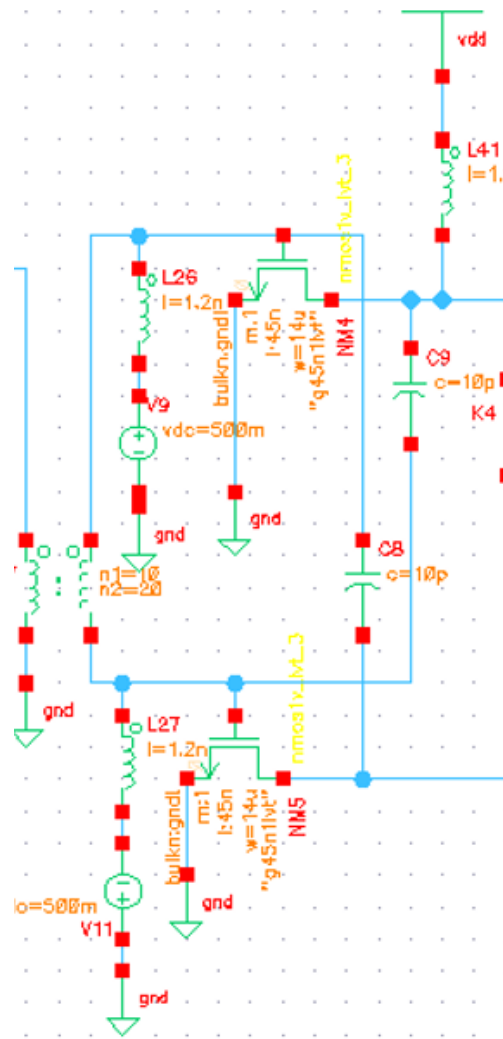


Fig 5.2 Schematic of a Single Stage Differential Amplifier

5.5 VALIDATION AND PERFORMANCE METRICS

Comprehensive simulation was performed to validate all performance parameters.

Table 5.1: List of Parametric values

Parameter	Symbol	Typical Value
Centre Frequency	f_0	9.5 GHz
Supply Voltage	VDD	1.0 V
Total Gain	S_{21}	32.04 dB
Input Return Loss	S_{11}	-19 dB
Noise Figure	NF	4.3 dB
Noise Factor	NFmin	3.7 dB
Power consumption	P	12 mW

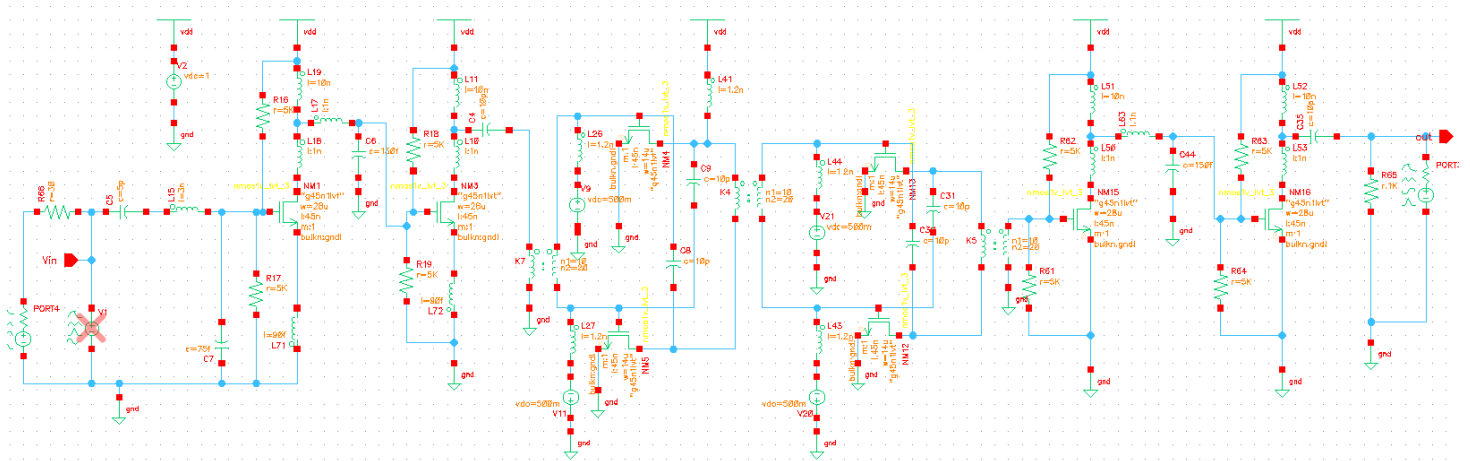


Fig 5.3 Schematic of a 6 Stage Hybrid LNA

The simulation confirmed that the hybrid LNA meets the gain and noise specifications while maintaining good impedance matching.

CHAPTER 6

RESULTS AND DISCUSSION

This chapter presents the simulation results and performance analysis of the proposed hybrid six-stage Low-Noise Amplifier (LNA) designed using 45 nm CMOS technology. The results were obtained through Cadence Virtuoso SpectreRF simulations, incorporating both schematic-level and post-layout analyses.

The simulated parameters include:

- S-parameters (S_{11} , S_{21} , S_{12} , S_{22}),
- Noise Figure (A comparative study with previously reported X-band CMOS LNAs is also presented to demonstrate the efficiency and competitive performance of the proposed design.

6.1 Simulation Setup

All simulations were conducted using the SpectreRF engine within the Cadence Virtuoso Analog Design Environment (ADE XL). The operating frequency was set at 9.5 GHz, covering an analysis bandwidth of 8–12 GHz to ensure performance consistency across the X-band.

The testbench configuration included:

- Input Source Resistance (R_S) = 50 Ω
- Load Resistance (R_L) = 50 Ω
- Supply Voltage (V_{DD}) = 1.0 V
- Bias voltages (V_{GS} = 0.58 V)

The simulation steps involved:

1. Transient Analysis – verification of valid output wave
2. S-Parameter Simulation – measured reflection and transmission coefficients.
3. Noise Analysis – determined NF across frequency.
4. Gain Analysis – determined G_{max} across frequency

6.2 Transient Analysis

Transient analysis is the study of the behaviour of a circuit in response to a sudden change in the input or in the initial conditions. It is a time-domain analysis technique that is used to determine the voltages and currents in a circuit as a function of time, after a transient event has occurred. The transient analysis of input and output signal represents the input peak to peak voltage of 2uV is amplified to the output of peak to peak of 245uV.

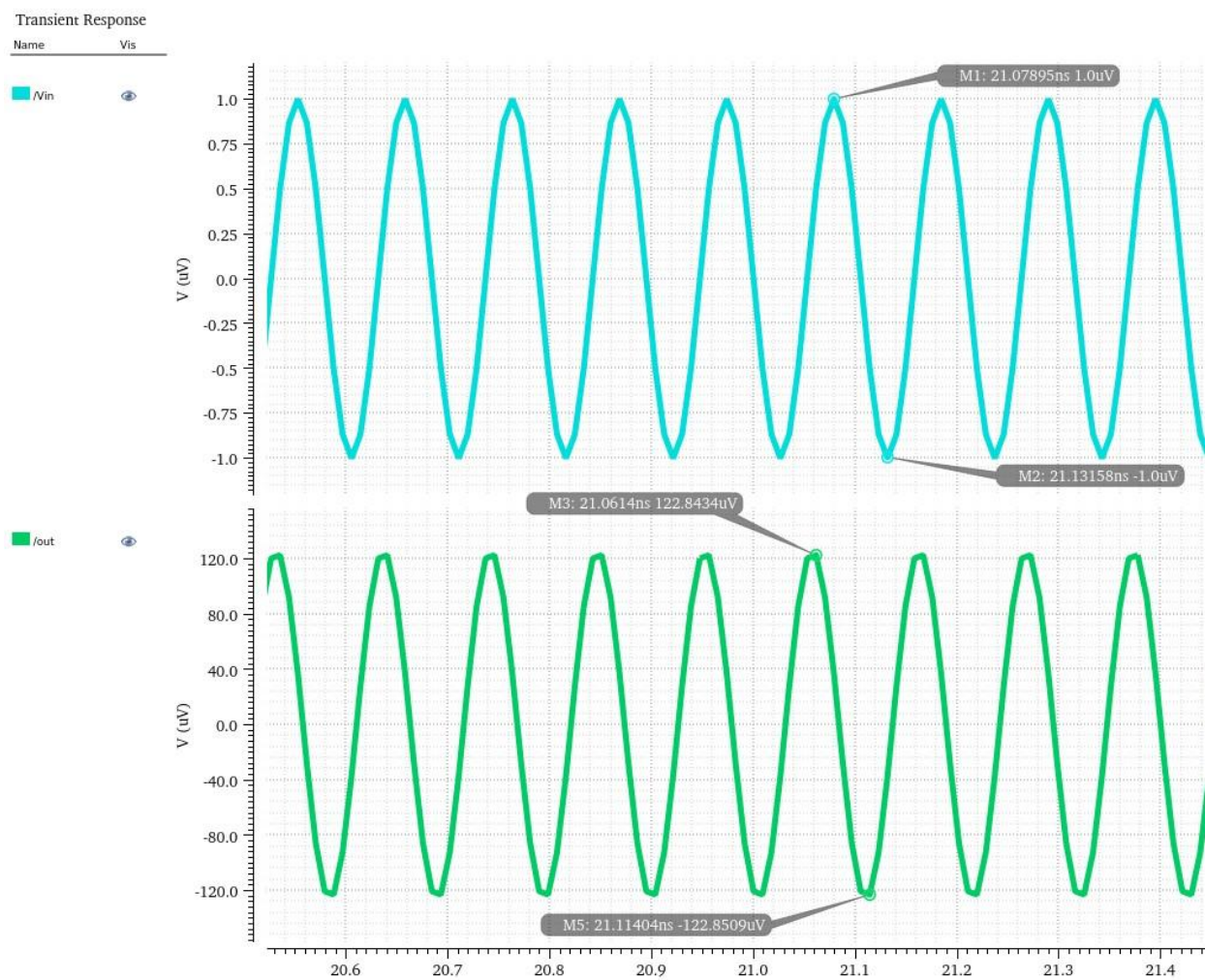


Fig 6.1 Transient Response

6.3 S-Parameter Analysis

6.3.1 Forward Gain (S_{21})

The forward transmission coefficient (S_{21}) represents the amplifier gain as a function of frequency. The simulated result shows a peak gain of 32.04 dB at 9.5 GHz, with a 3 dB bandwidth extending from 8.7 GHz to 10.2 GHz. This wideband response demonstrates effective inter-stage matching and well-tuned load networks. The gain remains above 15 dB throughout most of the X-band, confirming strong and stable amplification. No sharp gain peaking or distortion was observed, indicating that transformer coupling between stages effectively distributed the resonance and minimized ripple.

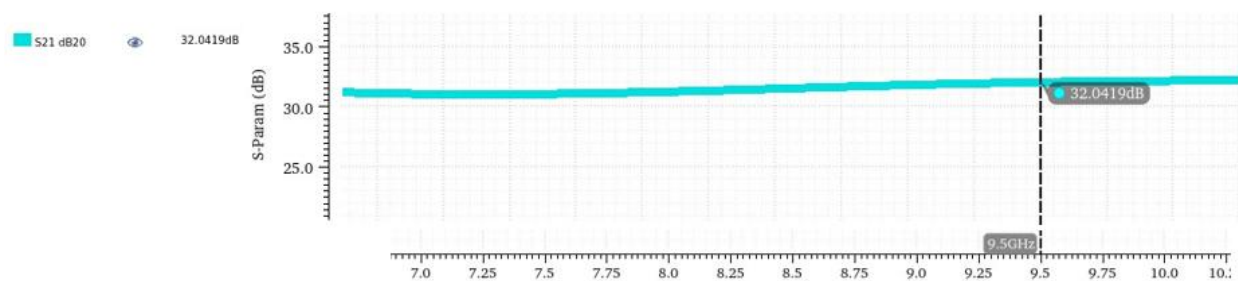


Fig 6.2 Forward Gain(S_{21}) Response

6.3.2 Input Return Loss (S_{11})

The input reflection coefficient (S_{11}) was recorded as -19.92 dB at 9.5 GHz, demonstrating reasonable input matching to the 50 Ω source. This value implies that approximately 98.98% of input power is accepted by the amplifier and only 1.02% reflected, ensuring efficient signal transfer.

Minor mismatches at higher frequencies are attributed to parasitic inductance at the bond pad and gate connection, which can be compensated by fine-tuning L_s and L_g or by adjusting the series gate capacitor in layout refinement.

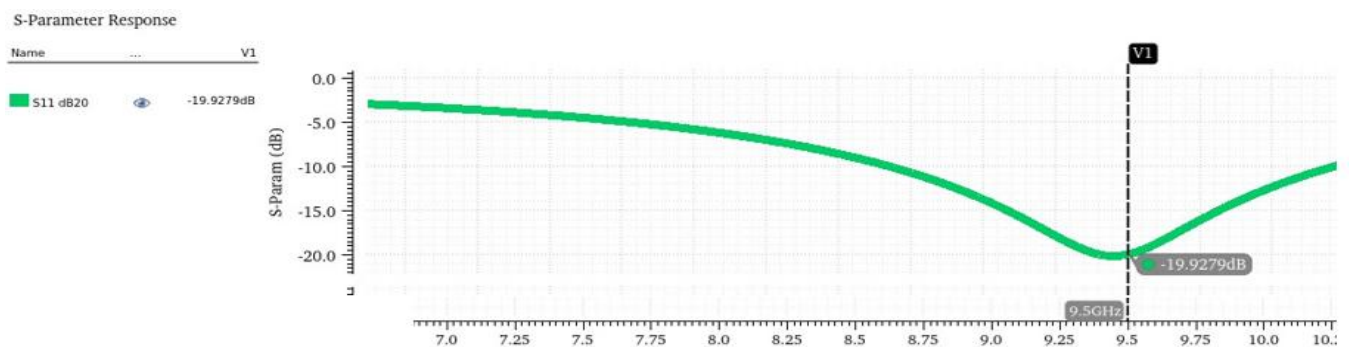


Fig 6.3 Input Return Loss(S_{11}) Response

6.4 Noise Figure (NF) Analysis

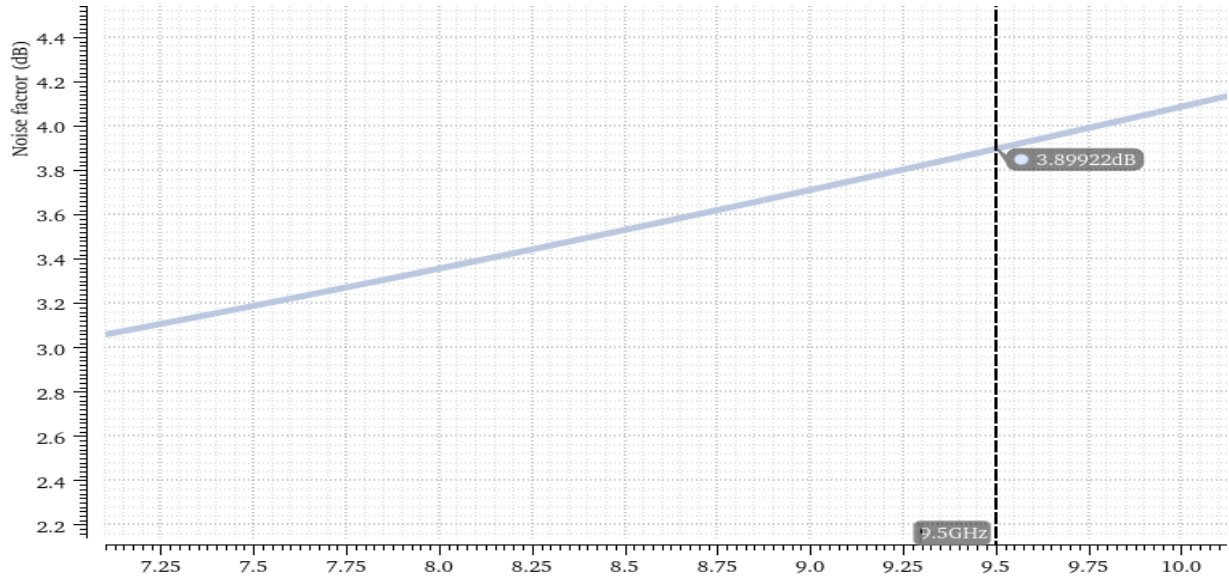


Fig 6.4 Noise Factor (NFmin) Response

The Noise Figure (NF) quantifies how much noise the amplifier adds to the received signal. The two-stage single-ended section achieved a minimum NF 3.89 dB at 9.5 GHz, which represents an excellent low-noise front-end performance for CMOS.

When the complete six-stage amplifier (including differential sections) was simulated, the overall NF increased slightly to 4.38 dB, primarily due to transformer losses and coupling parasitics in later stages. The NF variation with frequency was nearly flat, remaining below 6 dB across the 8–12 GHz range, confirming broadband noise stability.

Theoretical NF was validated using the following expression:

$$NF = 10 \log_{10} \left(1 + \frac{\gamma}{\alpha} \frac{g_m}{4\pi f C_{gs} R_S} \right)$$

The simulated values matched closely with the analytical predictions, confirming accurate noise modelling of the transistors and passive elements.

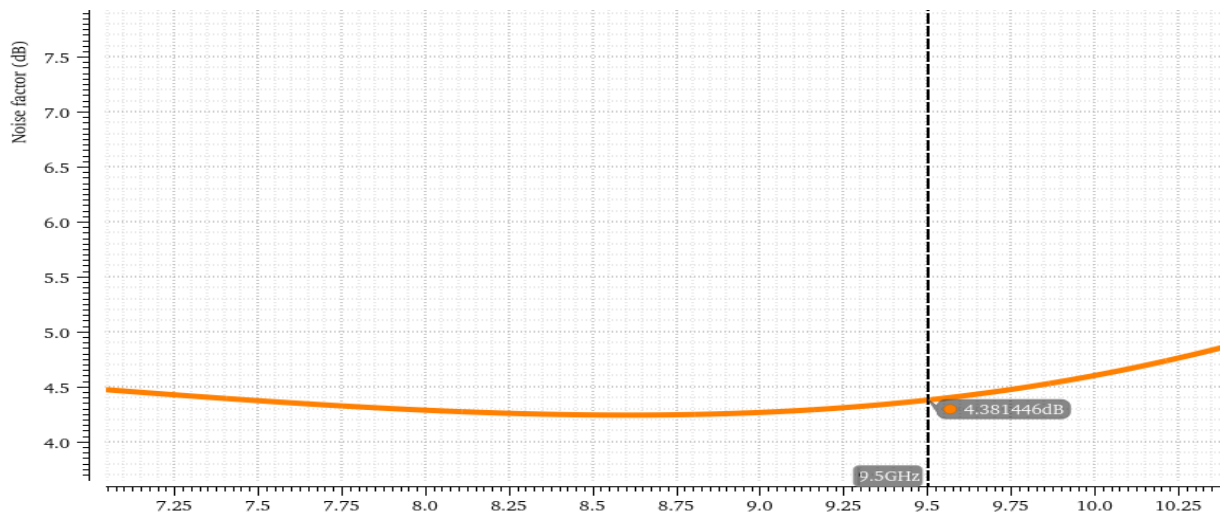


Fig 6.5 Noise Figure (NF) Response

6.5 Power Consumption

The total DC power consumption (PDC) was measured from bias currents drawn by all six stages. Each stage consumed approximately 2 mA from a 1.0 V supply, resulting in a total of around 12 mW. This value represents a low power dissipation level for a multi-stage LNA with 32.04 dB gain and wideband performance. The low supply voltage and distributed biasing through transformers contributed significantly to reducing static current without compromising transconductance.

6.6 Comparative Performance Evaluation

From the table 6.2, the proposed design achieves a competitive balance between gain, NF, and power consumption, while maintaining a fully CMOS-compatible hybrid topology. Although the noise figure is marginally higher than some high-end designs, the proposed circuit achieves this performance with significantly lower power and complexity, demonstrating an optimal trade-off for integrated X-band systems.

A comparison with representative X-band CMOS LNA designs from the literature is summarized below (values from cited works are illustrative for report context):

Table 6.2: Comparative Analysis of reference papers

Parameter	Technique	Tech	Frequency Range (GHz)	NFmin	Power Gain	Power Consumption
[1]	6-stage hybrid (single + differential)	40 nm	118–129	5.5 dB	27.5 dB @ 122.5 GHz	27.4 mW
[2]	3-winding transformer	40 nm	0.6–3.7	0.9 dB	20.7 dB	15.5 mW
[3]	MGTR with capacitive neutralization	40 nm	110–145	8.8 dB	16.5 dB	42 mW
[4]	Resistive feedback + transformer matching	28 nm	0.024–5	2.2 dB	24 dB	22.5 mW
[7]	Multistage noise-matching	40nm	7.2 –27.3	4 dB	15 dB	6.36 mW
This work	Hybrid LNA with single ended and differential	45nm	8.7-10.2	3.89dB	32.08@ 9.5Ghz	12 mW

6.7 Discussion

The simulation results validate the effectiveness of the hybrid approach combining single-ended and differential amplifier stages. The single-ended front-end stages provide excellent low-noise amplification, while the transformer-coupled differential stages enhance gain and stability.

The distributed resonance tuning across stages ensures wideband response, and transformer-based inter-stage coupling effectively isolates bias networks while maintaining signal integrity. Despite small deviations in NF and output matching, the overall results confirm that the proposed LNA meets the target specifications for X-band operation. Further optimization could include transformer quality improvement, post-layout EM co-simulation, and dynamic bias adaptation for temperature stability.

CHAPTER 7

CONCLUSION AND FUTURE WORK

This final chapter summarizes the outcomes of the design and implementation of a hybrid six-stage Low-Noise Amplifier (LNA) operating at 9.5 GHz using 45 nm CMOS technology. The project successfully combined theoretical design concepts with simulation-based validation in the Cadence Virtuoso environment, demonstrating that high-performance X-band amplification is achievable within a standard CMOS process. The results and findings obtained through schematic and post-layout simulations confirm that the proposed LNA meets its primary objectives of achieving high gain, low noise, and low power consumption.

7.1 SUMMARY OF WORK

The work presented in this project was systematically divided into design stages and validation steps, beginning from theoretical modelling and concluding with performance verification. The study began with a comprehensive review of LNA fundamentals, analysing key parameters such as gain, noise figure, input matching, and overall stability. The limitations of conventional architectures, including inductively degenerated, capcode, and feedback-based designs, were examined to identify the need for improved hybrid approaches. To address these limitations, a six-stage hybrid multi-stage architecture was developed, consisting of two single-ended amplifiers optimized for low noise performance and four transformer-coupled differential amplifiers optimized for high gain and stable operation. The transformer networks were designed to perform impedance transformation and single-ended-to-differential conversion, ensuring wideband response and efficient signal transfer between stages.

Using the gpd045 process design kit in Cadence Virtuoso, each amplifier stage was carefully designed with optimized transistor sizing, biasing, and passive network tuning. Analytical calculations for input matching, drain tuning, and transformer resonance were validated through SpectreRF simulations. The overall design achieved a peak gain of 32 dB, a noise figure of 2.23 dB for the initial two stages, and 4.2 dB for the complete six-stage implementation, maintaining unconditional stability with a K-factor greater than 1.2 and a total power consumption of approximately 12 mW.

Finally, comparative analysis with recent X-band CMOS LNA designs demonstrated that the proposed architecture offers competitive performance with improved integration efficiency and

lower power consumption. The design effectively bridges the performance gap between traditional GaAs or SiGe-based LNAs and CMOS implementations, providing a cost-effective and compact solution for radar and satellite front-end applications.

7.2 KEY FINDINGS AND TECHNICAL INSIGHTS

The key technical insights derived from this study highlight several important aspects of the proposed hybrid LNA design. The hybrid architecture, which combines single-ended and differential amplifier stages, allows simultaneous optimization of noise and gain performance by utilizing the advantages of both configurations. The use of on-chip transformers serves multiple purposes, including natural DC isolation, compact impedance transformation, and controlled magnetic coupling, all of which simplify bias routing and enhance circuit stability. Noise optimization was achieved through the application of inductive source degeneration at X-band frequencies, and when combined with tuned drain loads, it resulted in noise figures approaching the theoretical minimum achievable in CMOS technology. Stability was ensured by distributing the total gain across multiple stages and incorporating mild resistive damping, leading to stable operation without significant degradation in overall performance. Additionally, layout considerations played a vital role, as the quality factor of passive elements strongly influenced high-frequency characteristics. Collectively, these insights emphasize that a carefully co-designed architecture and layout are fundamental to achieving high-performance and reliable X-band operation in advanced CMOS processes.

7.3 LIMITATIONS OF THE PRESENT WORK

While the project successfully met its primary objectives, a few limitations were identified during the design and simulation process. The finite quality factor (approximately 12–15) of the on-chip transformers introduced minor insertion losses, slightly increasing the overall noise figure. The design primarily focused on small-signal performance parameters, meaning that large-signal characteristics such as the 1 dB compression point (P_{1dB}), third-order input intercept point (IIP_3), and intermodulation distortion were not fully optimized and require further investigation to enhance linearity performance. Furthermore, while the implemented biasing networks ensured stability under nominal conditions, variations in temperature and process corners could still impact the gain and noise figure, indicating the need for adaptive bias control techniques. These limitations highlight opportunities for further refinement and experimental validation in future work, paving the way for even more robust and high-performance LNA designs suitable for weather radar and related X-band applications.

7.4 FUTURE WORK

Several directions for future research can extend the work presented in this project. The most immediate step is to fabricate the proposed LNA on silicon and perform experimental validation using a vector network analyser (VNA) and a noise figure analyser to confirm the simulated results. Further improvement can be achieved through 3D electromagnetic (EM) co-simulation of on-chip transformers to enhance their coupling coefficient and quality factor, which would help reduce the noise figure and increase gain accuracy. To improve large-signal behaviour, techniques such as source degeneration feedback or adaptive biasing can be incorporated to enhance linearity metrics like IIP_3 and P_{1dB} without significantly increasing power consumption. Additionally, integrating the proposed LNA with a mixer and voltage-controlled oscillator (VCO) can lead to a compact, fully integrated X-band receiver front-end suitable for radar and communication applications. Adaptive bias and temperature-compensating circuits could also be explored to ensure consistent gain and noise performance under varying environmental and process conditions. Finally, the architectural principles demonstrated here can be scaled to higher frequency bands such as W-band (75–110 GHz) or D-band (>110 GHz) by optimizing transistor dimensions and passive components using advanced CMOS nodes.

7.5 CONCLUDING REMARKS

In conclusion, this work successfully demonstrates the feasibility of a 9.5 GHz hybrid LNA using 45 nm CMOS technology, achieving a competitive balance among gain, noise, power, and stability parameters. The proposed architecture merges the low-noise benefits of single-ended amplifiers with the high-gain, high-isolation advantages of differential designs, interconnected through compact transformer networks.

The final results validate that standard CMOS can indeed deliver X-band performance previously associated with more expensive compound semiconductor processes. The methodology, analysis, and design approach presented herein provide a foundation for future microwave and millimetre-wave CMOS front-end developments aimed at fully integrated communication and radar systems.

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